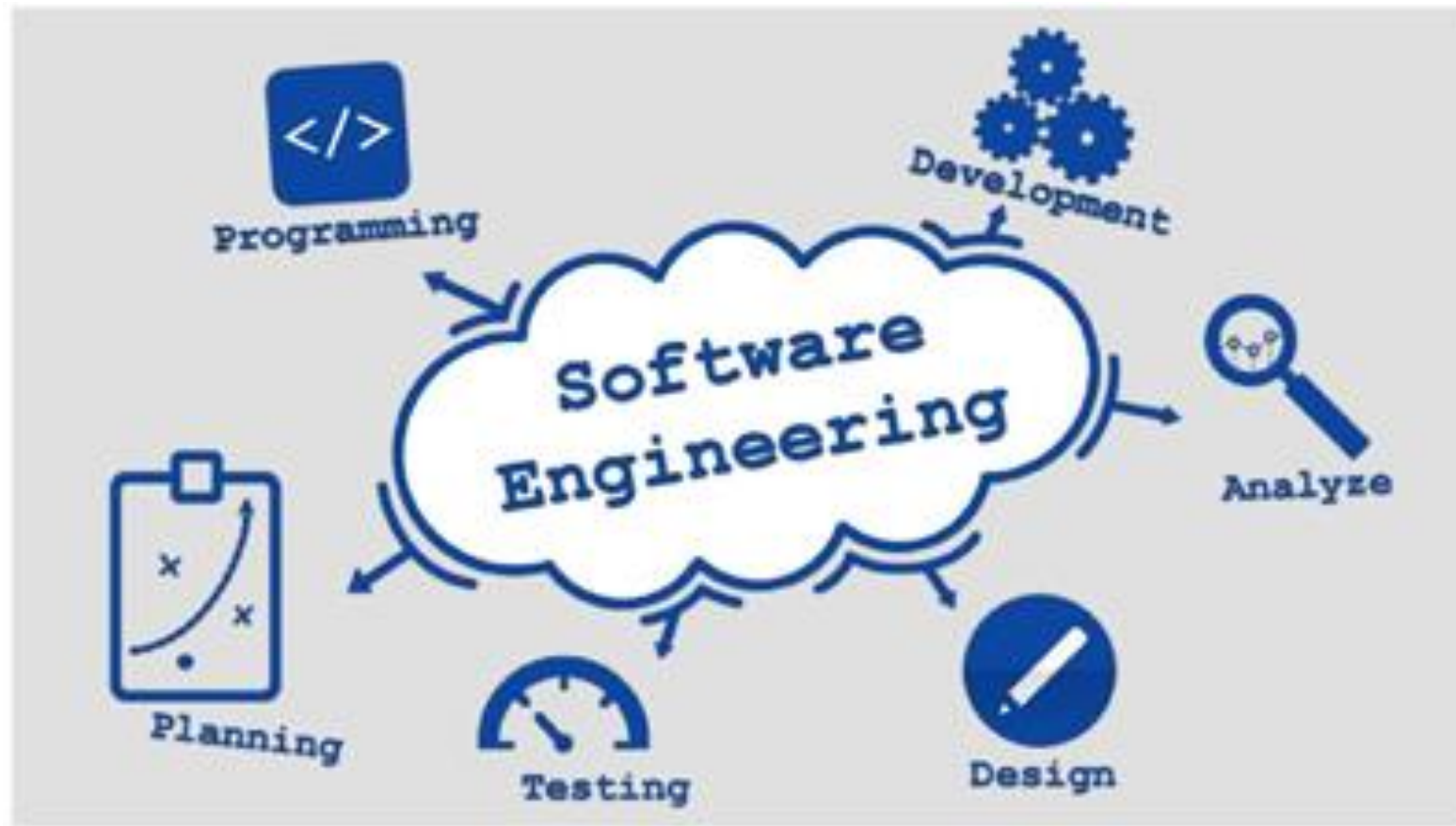




SOFTWARE ENGINEERING



Architecture

System Architecture Styles

- Monolith
- Modular Monolith
- Microservices
- Microkernel



Architecture

Communication Styles

- Client-Server
- Event-driven
- Peer-to-Peer



Architecture

Application Internal Patterns (Architecture Patterns)

- Layered
- MVC / MVP / MVVM
- CQRS

Data Processing / Flow Architectures

- Pipe-Filter



Architecture

A real system usually looks like:

- Client-Server + Microservices + Event-driven + Layered + MVC



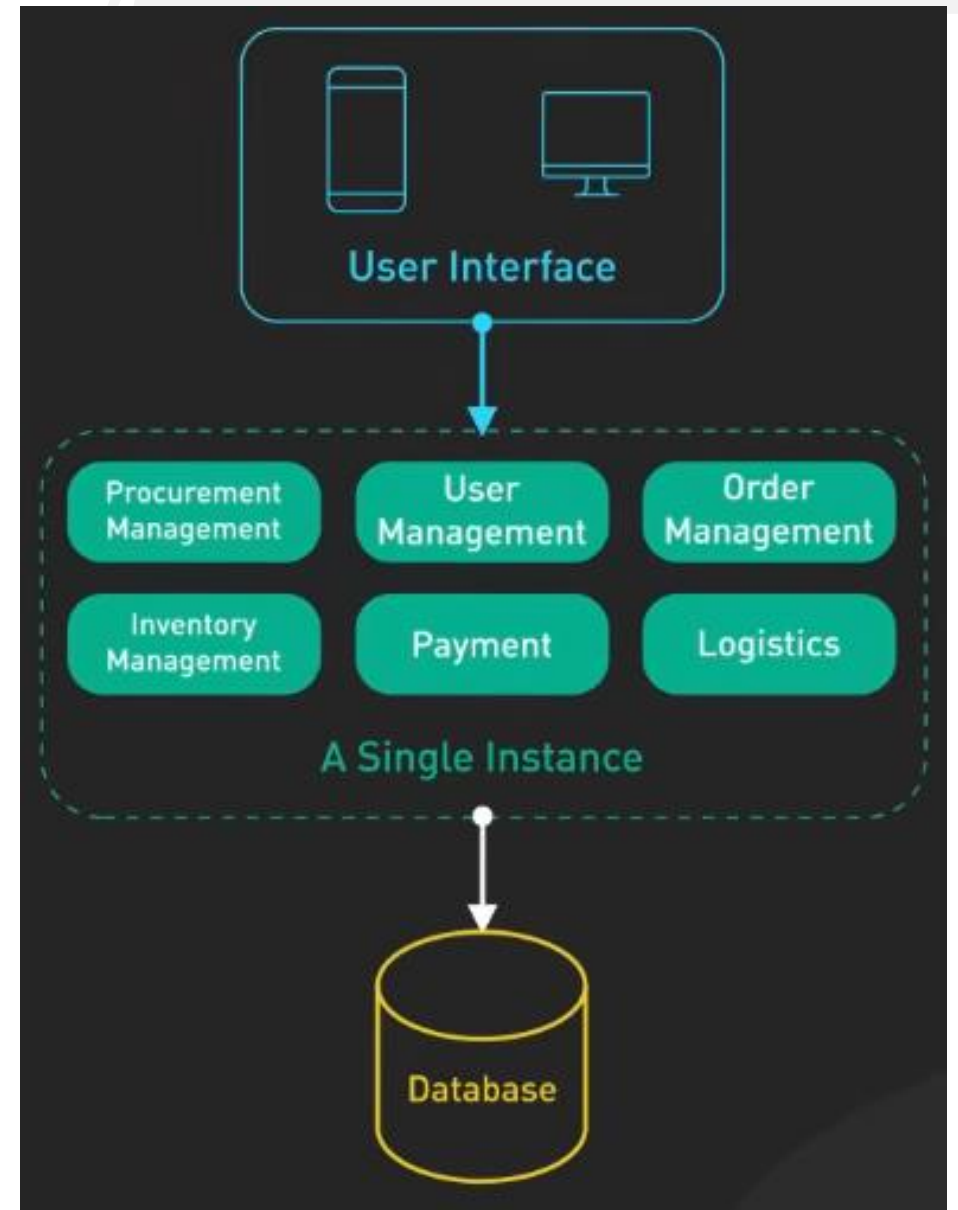
Architecture

System Architecture Styles



Architecture

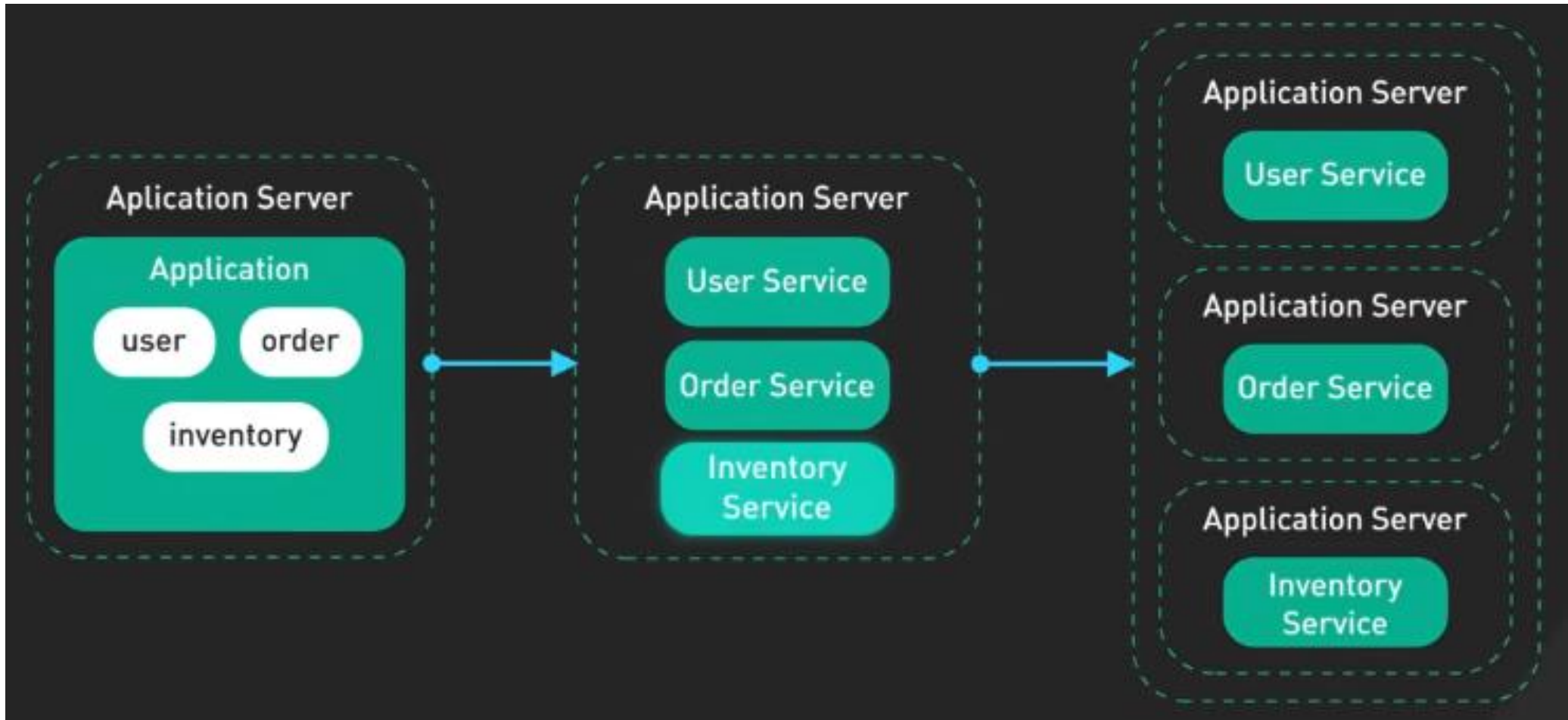
- Monolithic Architecture
- 1 project has everything (UI, Business Logic,)





Architecture

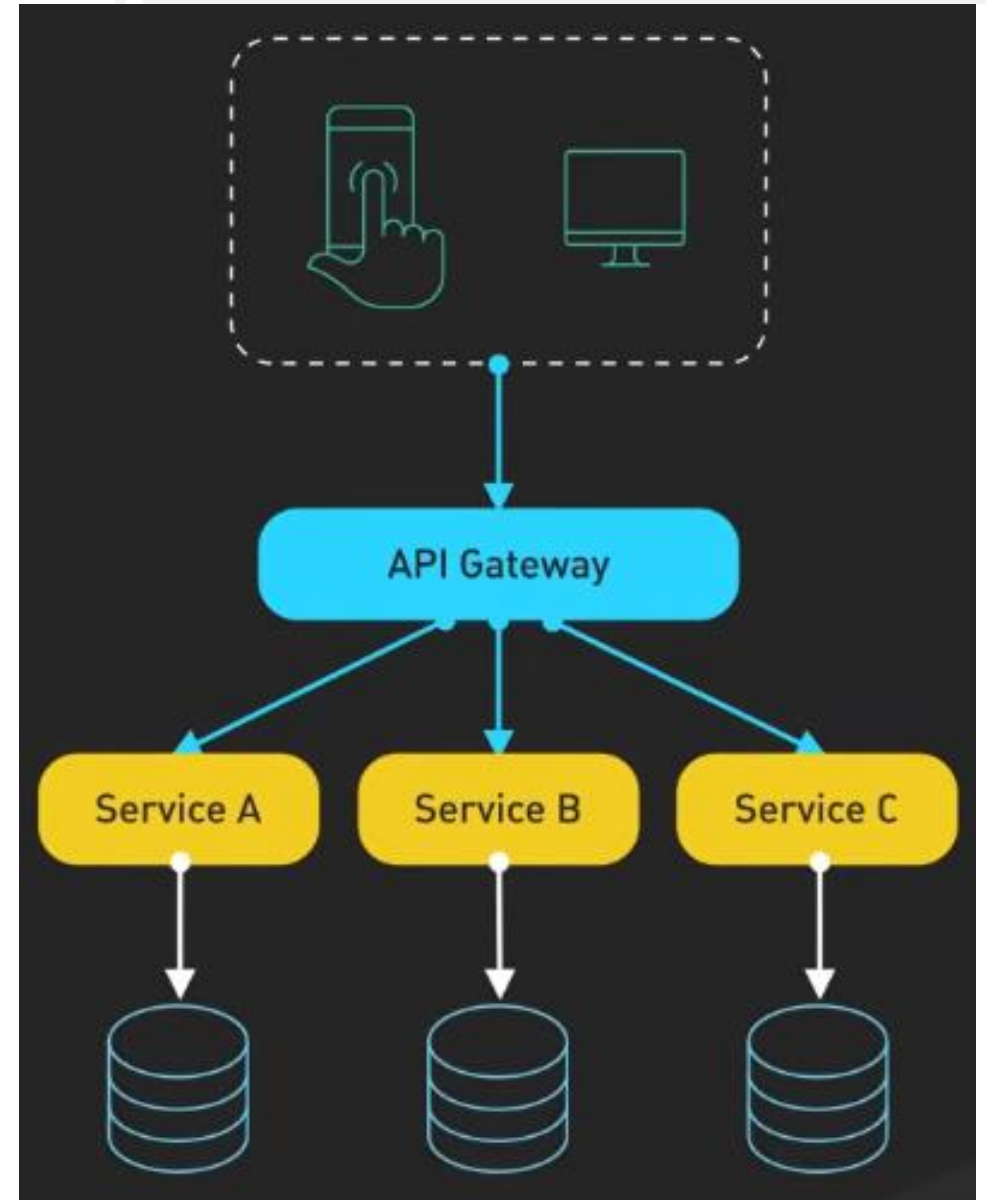
- Modular Monolithic Architecture (The middle Figure)
- On the left: Monolithic, on the right: Microservice





Architecture

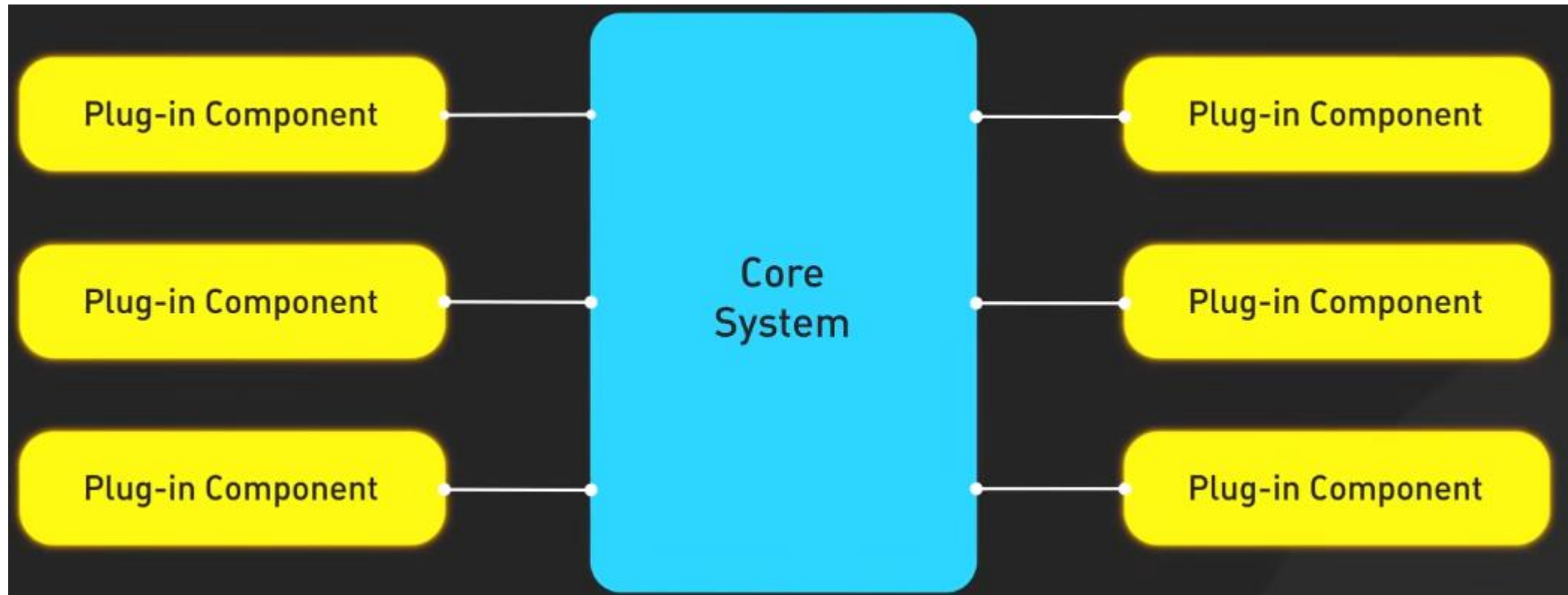
- Microservice Architecture
- Split app into many separated apps/services





Architecture

- Microkernel Architecture





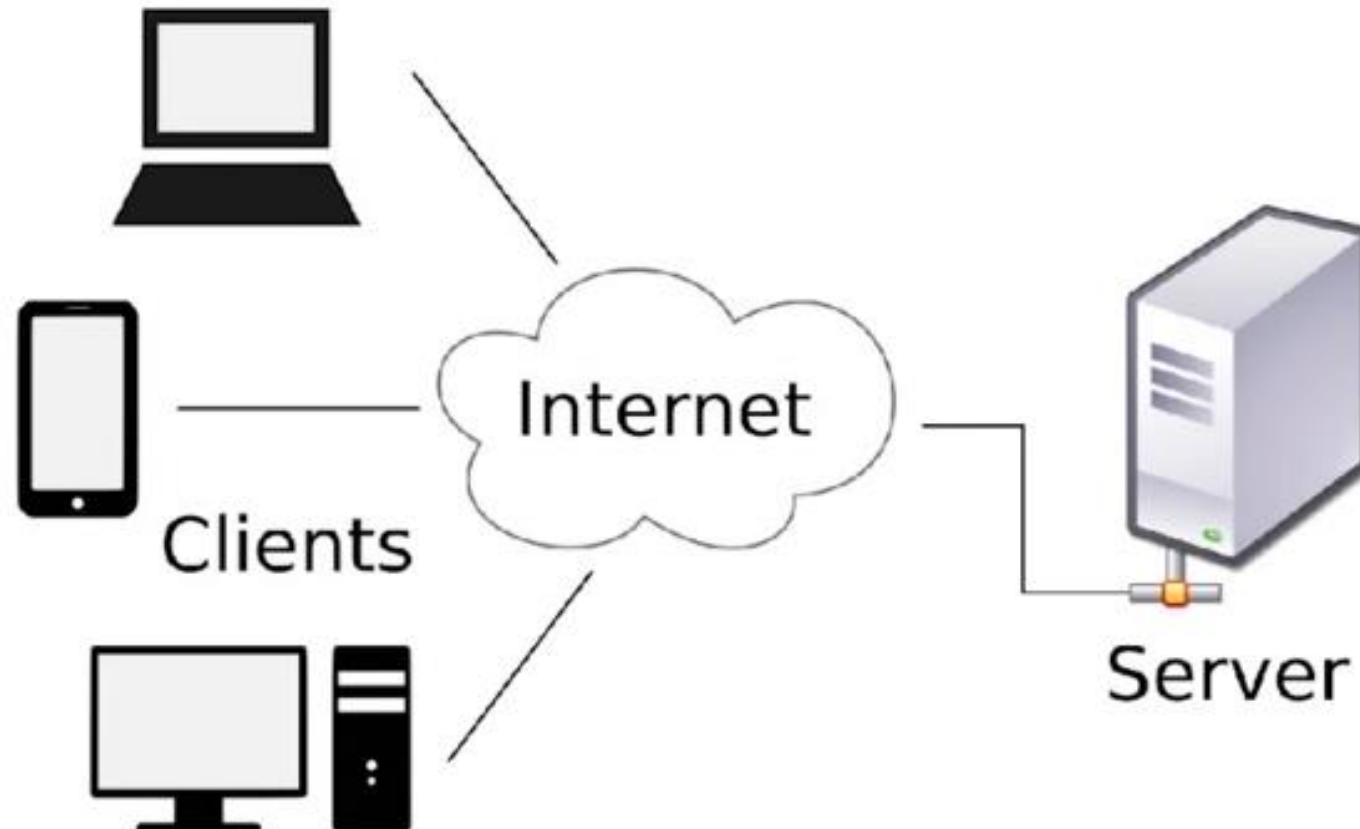
Architecture

Communication Styles



Architecture

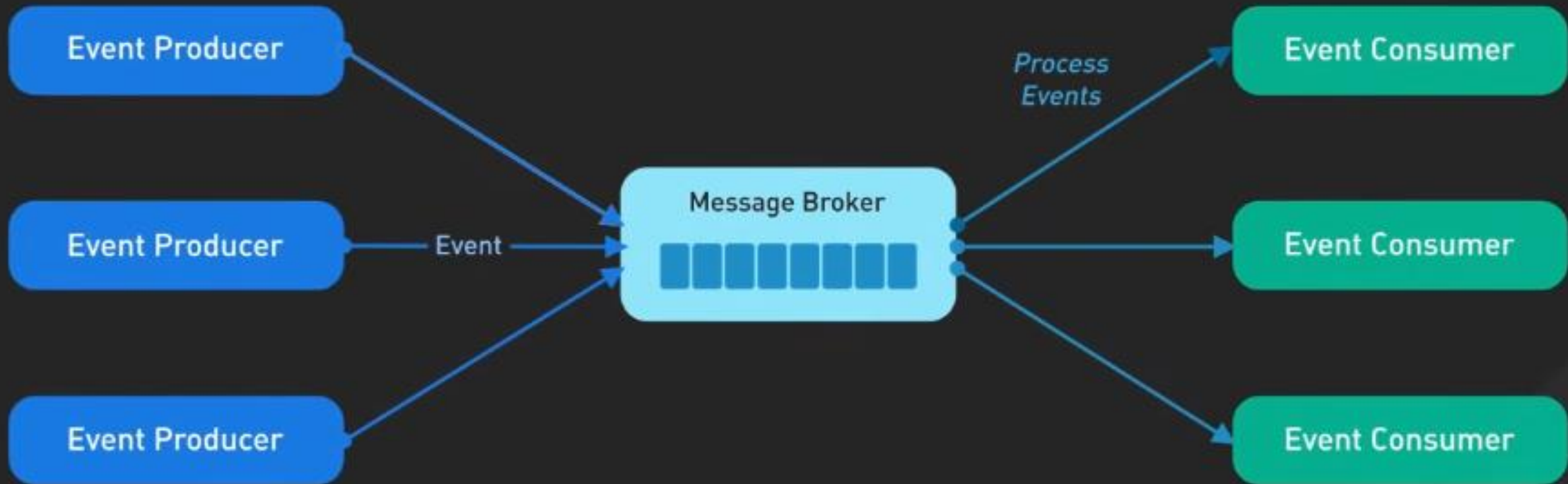
- Client-Server Architecture (Most common)





Architecture

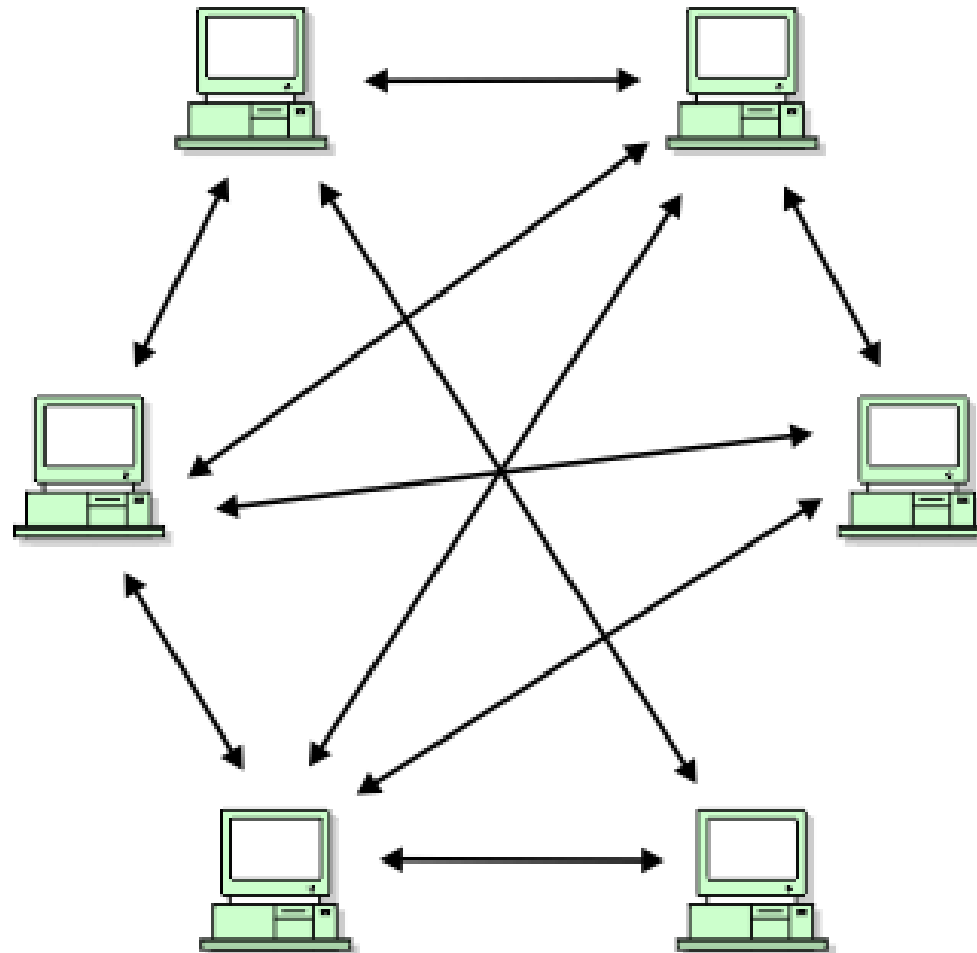
- Event-driven Architecture
- Used as a way of communication between services (Apps) in Microservice





Architecture

- Peer-to-Peer Architecture





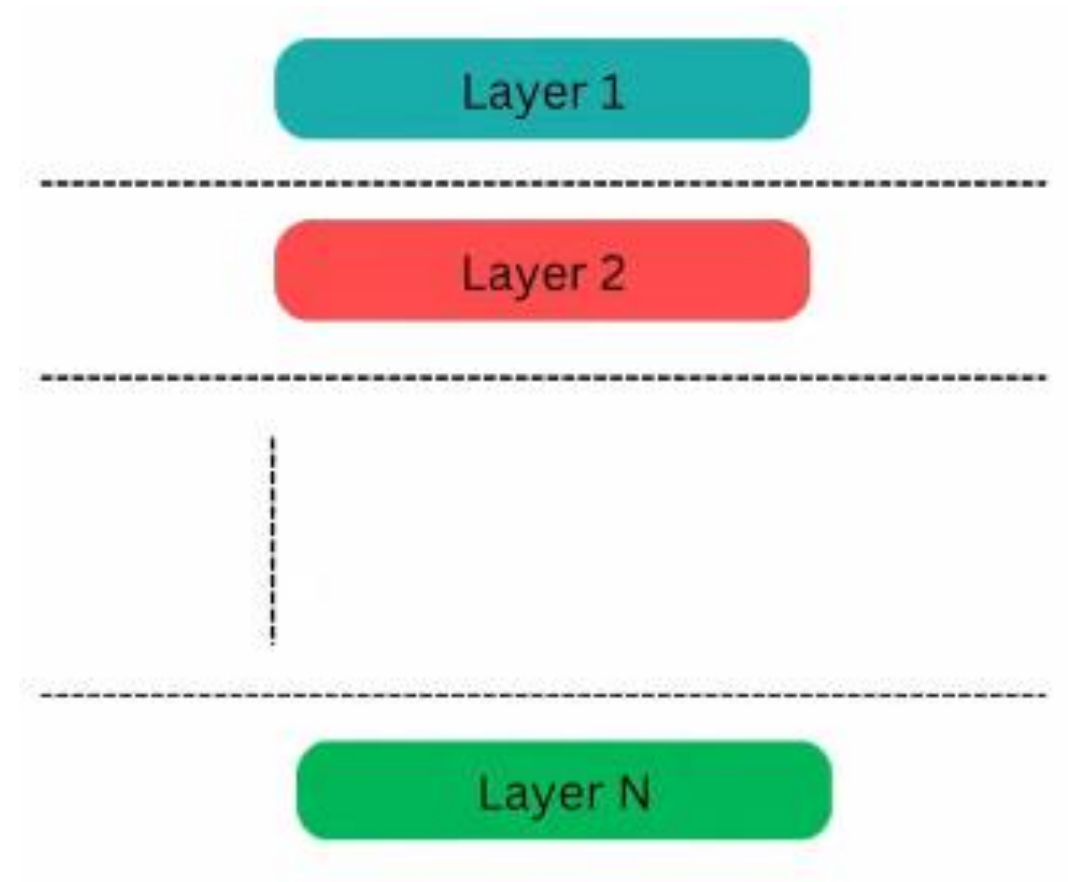
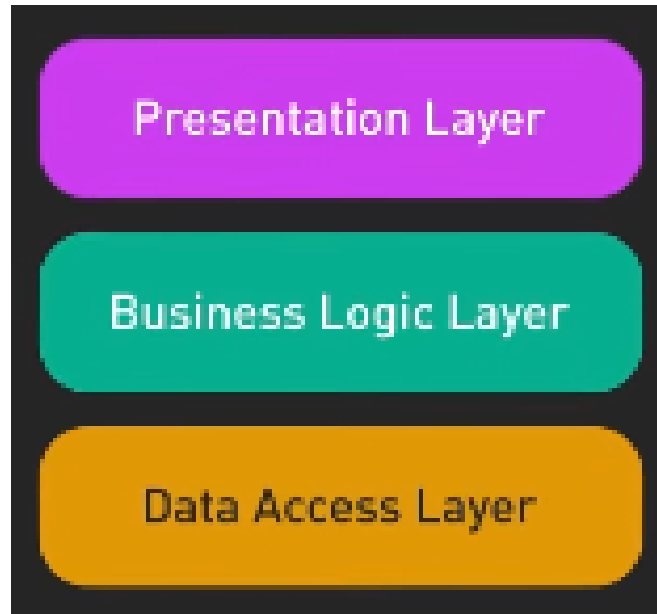
Architecture

Application Internal Patterns (Architecture Patterns)



Architecture

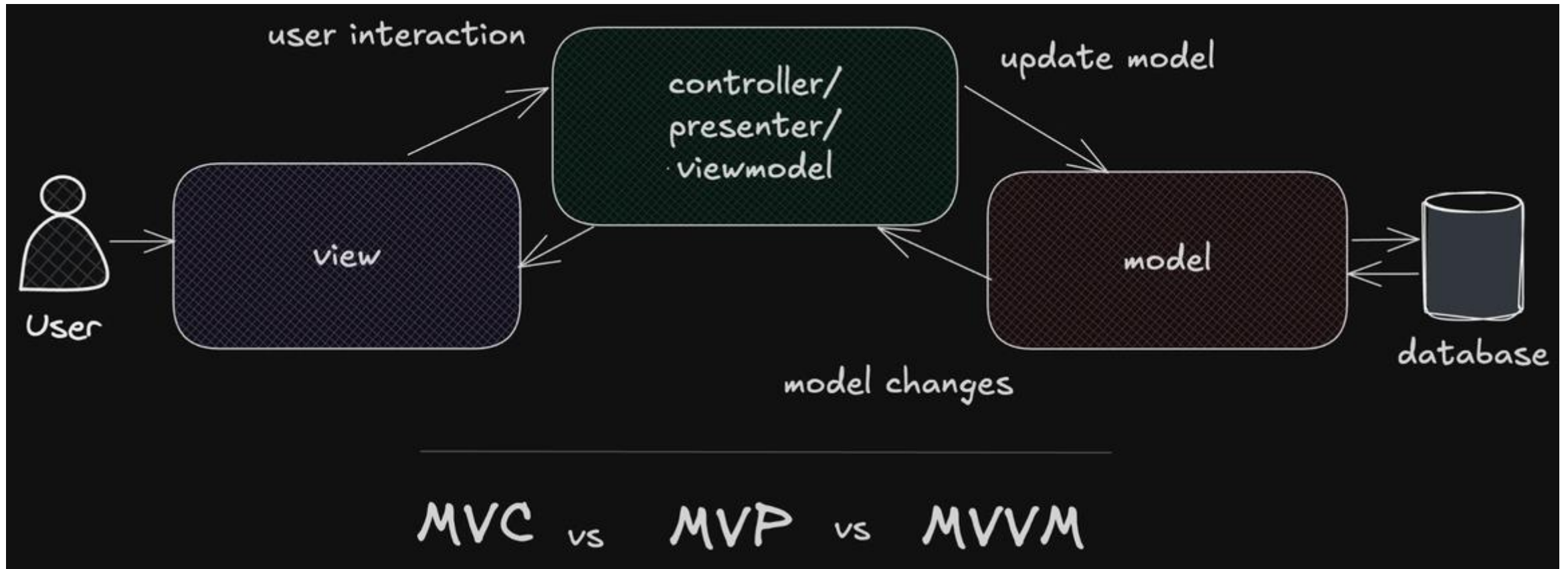
- Layered Architecture (such as 3-Tier)





Architecture

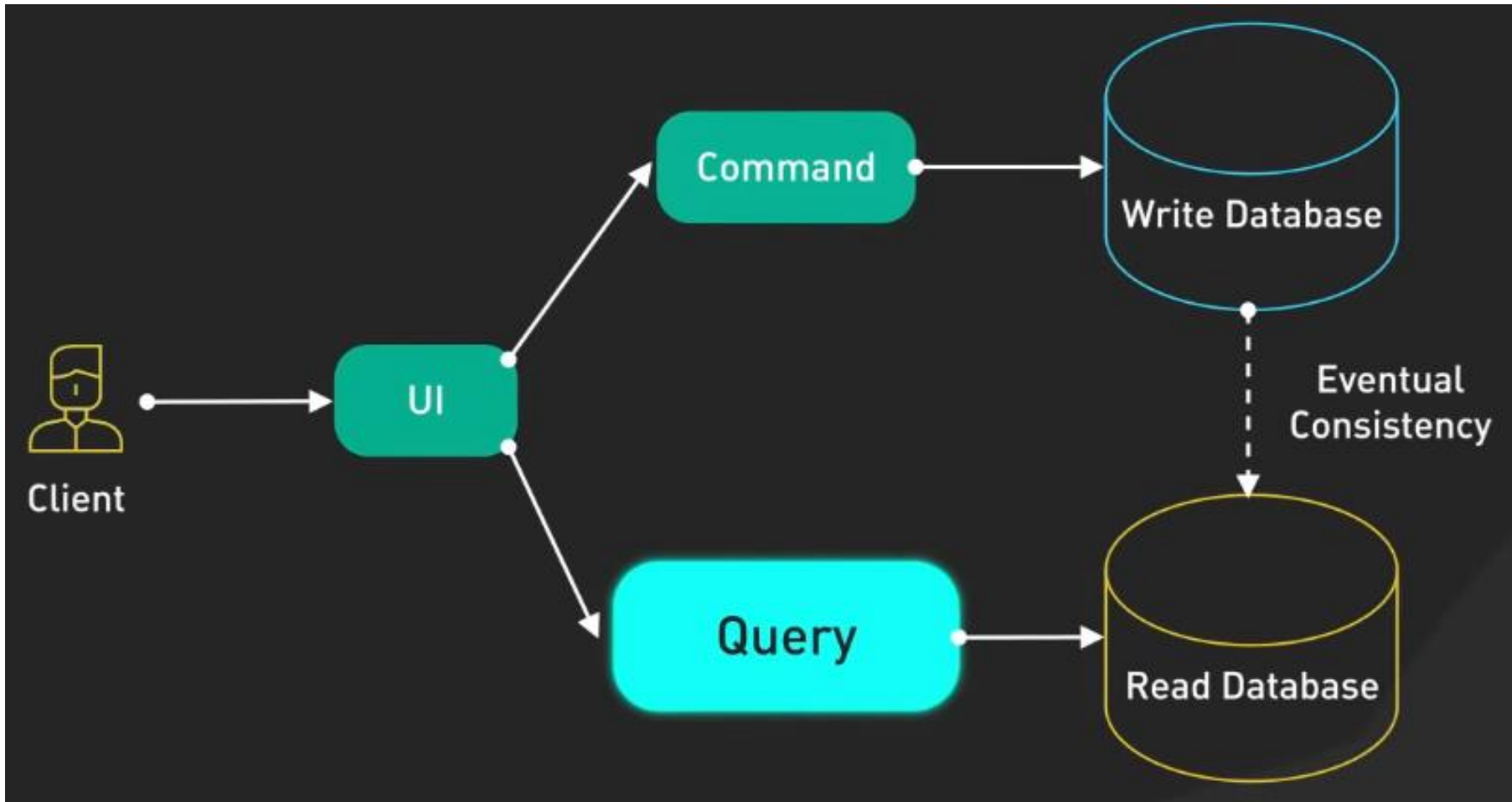
- MVC vs MVP vs MVVM





Architecture

- Command Query Responsibility Segregation (CQRS)
- Used in Microservice Architectures (a way of communication using Event-Driven)





Architecture

Data Processing / Flow Architectures



Architecture

- Pipe-Filter Architecture
- Used in ETL Systems (Data Pipelines)

